Power Systems Analysis & Electrical Machines

Department: Electrical Engineering | Semester: 5 | Code: EE501
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1. Transmission Line Models & Performance

Transmission lines are modeled as Short (<80 km), Medium (80-250 km, Nominal Pi/T models), and Long (>250 km, distributed parameters via wave propagation constants).

2. Symmetrical Components & Fault Analysis

Any unbalanced three-phase system can be decomposed into Positive (V1), Negative (V2), and Zero (V0) sequence symmetrical components using Fortescue's theorem.

3. Synchronous Machines & Stability

Swing equation governs rotor dynamics during disturbances: (H / pi*f0) * (d^2 delta / dt^2) = P_mech - P_elec.